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# Road Object Detection in Fisheye Camera

*A Comparative Evaluation of Nine YOLO Model Variants on the FishEye8K Benchmark*

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# Why Road Object Detection Matters

1

### Autonomous Driving

Real-time detection of vehicles, pedestrians and cyclists enables fast, safe decision-making.

2

### Traffic Management

Accurate object counts support congestion monitoring and smarter, more responsive cities.

3

### The YOLO Family

YOLOv9, v10, v11 and v12 continue to push the speed/accuracy balance for real-time detection.

### The Fisheye Challenge

Fisheye lenses cover 180°+ field of view, producing strong curvilinear distortion — objects near image edges appear elongated or bent, while scale varies sharply between near and far objects. Standard detectors trained on undistorted imagery struggle to generalize to this geometry.

# Research Objective

Compare the performance of nine YOLO model variants (v9, v9-SE, v10, v11, v12) for detecting road objects on the FishEye8K dataset, evaluating accuracy, speed, and robustness to fisheye distortion — in order to identify the optimal architecture for real-time fisheye traffic surveillance.

- Train 9 model variants under identical conditions
- Evaluate mAP@0.5, mAP@0.5:0.95, Precision, Recall, F1
- Compare inference speed for real-time suitability
- Explore fine-tuning strategies to push accuracy further

### Model Generations Evaluated

**YOLOv9**

c · e · SE (attention)

**YOLOv10**

l · x

**YOLO11**

l · x

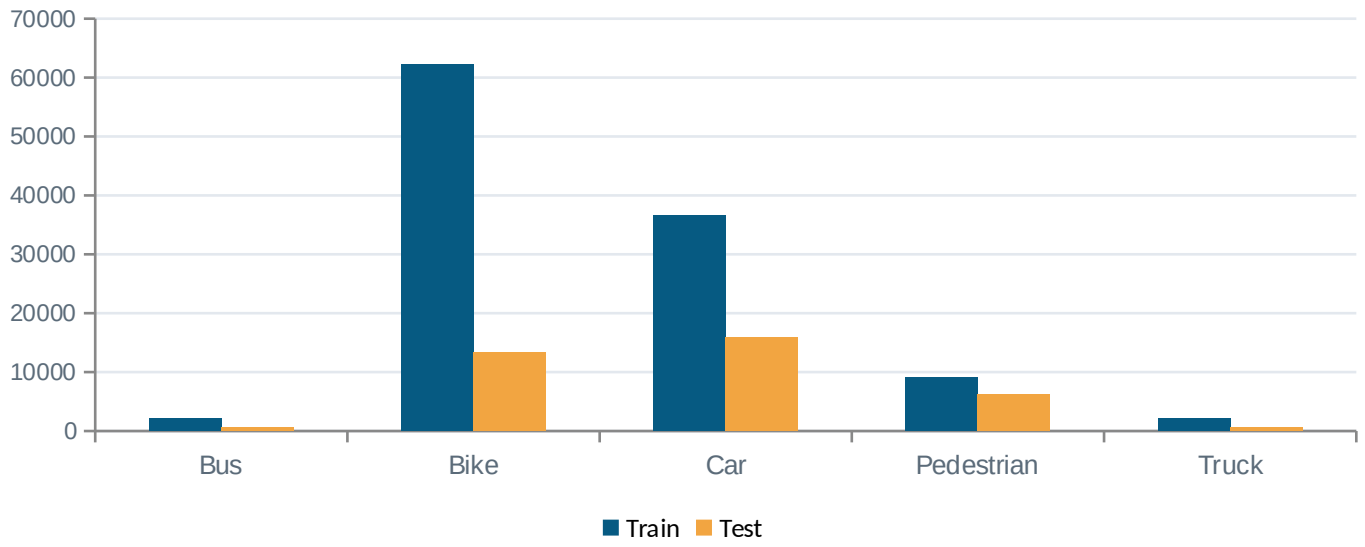
**YOLO12**

l · x (attention-centric)

# FishEye8K Benchmark Dataset

| 5,288 | 112,205 | 1,000 | 5 |
|---|---|---|---|
| training images | training instances | test images | object classes |

## Instance Distribution by Class (Train / Test)




### Why It's Hard

**Severe class imbalance:** Bike & Car dominate; Bus/Truck are rare.
**Wide scale range:** object width vs. height varies sharply across the frame.
**Radial layout:** objects cluster near the fisheye boundary and center.

# Nine Models, Four Generations

| | |
|---|---|
| **YOLOv9 (c / e)** | GELAN + Programmable Gradient Information; anchor-free head for efficient real-time detection. |
| **YOLOv9-SE** | YOLOv9 baseline + Squeeze-and-Excitation attention for channel-wise feature recalibration. |
| **YOLOv10 (l / x)** | NMS-free end-to-end detection via consistent dual assignment; spatial-channel decoupled downsampling. |
| **YOLO11 (l / x)** | C3k2 block + C2PSA (position-sensitive attention) for better multi-scale accuracy. |
| **YOLO12 (l / x)** | Attention-centric architecture (area attention + R-ELAN), departing from CNN-only design. |

# Training Configuration & Metrics

## Training Setup

| | |
|---|---|
| **Hardware** | NVIDIA A100-SXM4-40GB (Google Colab Pro) |
| **Input size** | 640 × 640 |
| **Epochs** | 100 (baseline) |
| **Batch size** | 16 |
| **Optimizer** | SGD, momentum 0.937 |
| **Learning rate** | 0.01 initial, weight decay 0.0005 |
| **Pretraining** | MS COCO |

## Evaluation Metrics

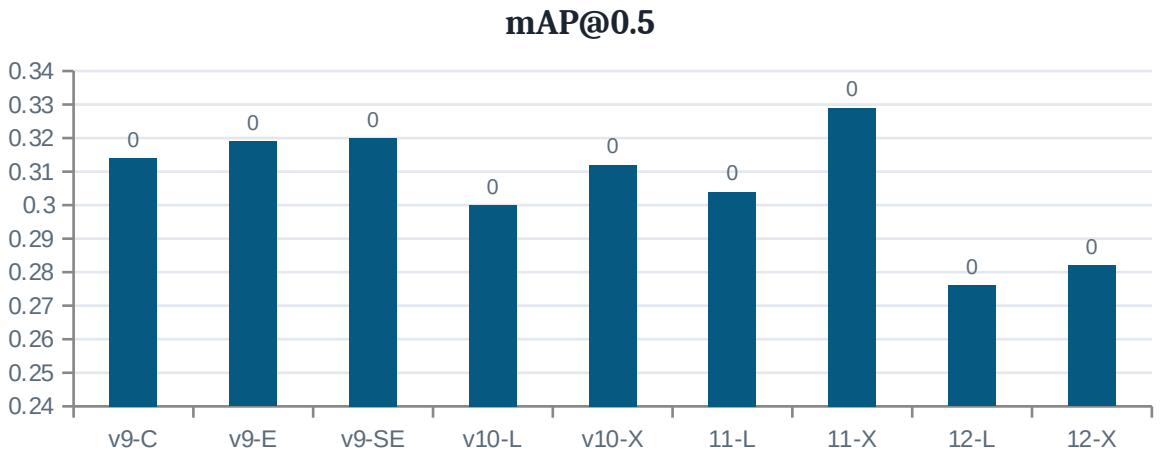
| | |
|---|---|
| **Precision** | TP / (TP + FP) — false-alarm rate |
| **Recall** | TP / (TP + FN) — miss rate |
| **F1-Score** | Harmonic mean of Precision & Recall |
| **mAP@0.5** | Mean AP at IoU threshold 0.5 |
| **mAP@0.5:0.95** | Mean AP averaged over IoU 0.5–0.95 |
| **Inference time** | ms per image — real-time suitability |

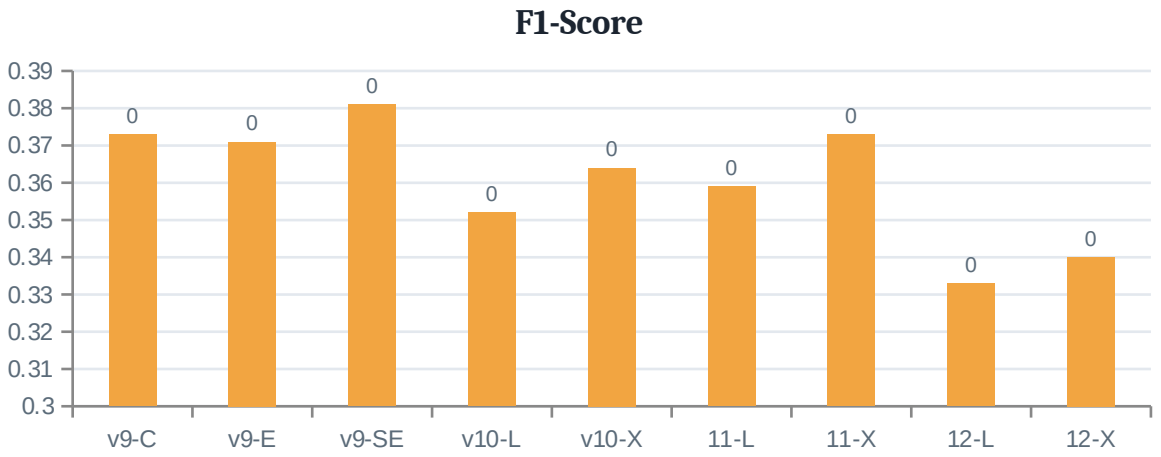
# Overall Performance — All Nine Models

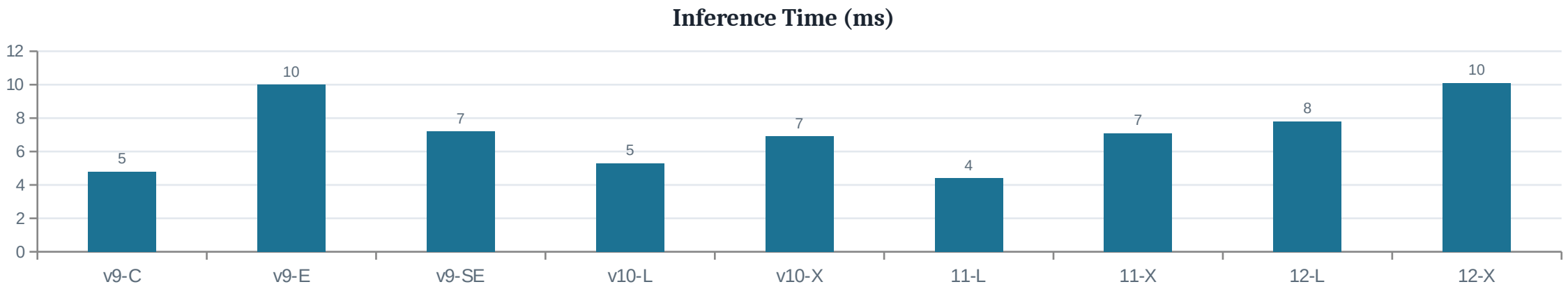
| Model | Precision | Recall | mAP@0.5 | mAP@.5:.95 | F1 | Infer (ms) |
|---|---|---|---|---|---|---|
| **YOLOv9-C** | 0.486 | 0.303 | 0.314 | 0.154 | 0.373 | 4.8 |
| **YOLOv9-E** | 0.455 | 0.313 | 0.319 | 0.156 | 0.371 | 10.0 |
| **YOLOv9-SE** | 0.474 | 0.318 | 0.320 | 0.150 | **0.381** | 7.2 |
| **YOLOv10-L** | 0.451 | 0.289 | 0.300 | 0.144 | 0.352 | 5.3 |
| **YOLOv10-X** | 0.507 | 0.284 | 0.312 | 0.156 | 0.364 | 6.9 |
| **YOLO11-L** | 0.453 | 0.297 | 0.304 | 0.147 | 0.359 | 4.4 |
| **YOLO11-X** | 0.502 | 0.297 | **0.329** | 0.166 | 0.373 | 7.1 |
| **YOLO12-L** | 0.443 | 0.267 | 0.276 | 0.132 | 0.333 | 7.8 |
| **YOLO12-X** | 0.445 | 0.276 | 0.282 | 0.137 | 0.340 | 10.1 |

*Highlighted: best mAP@0.5 (YOLO11-X) and best F1-Score (YOLOv9-SE)*

# Accuracy Comparison Across Models



# Speed vs. Accuracy Trade-off

Fastest

## YOLO11-L

4.4 ms inference — best fit for strict real-time latency budgets.

Most Accurate

## YOLO11-X

Highest mAP@0.5 (0.329) at a still-practical 7.1 ms — best for precision-critical deployments.

Most Balanced

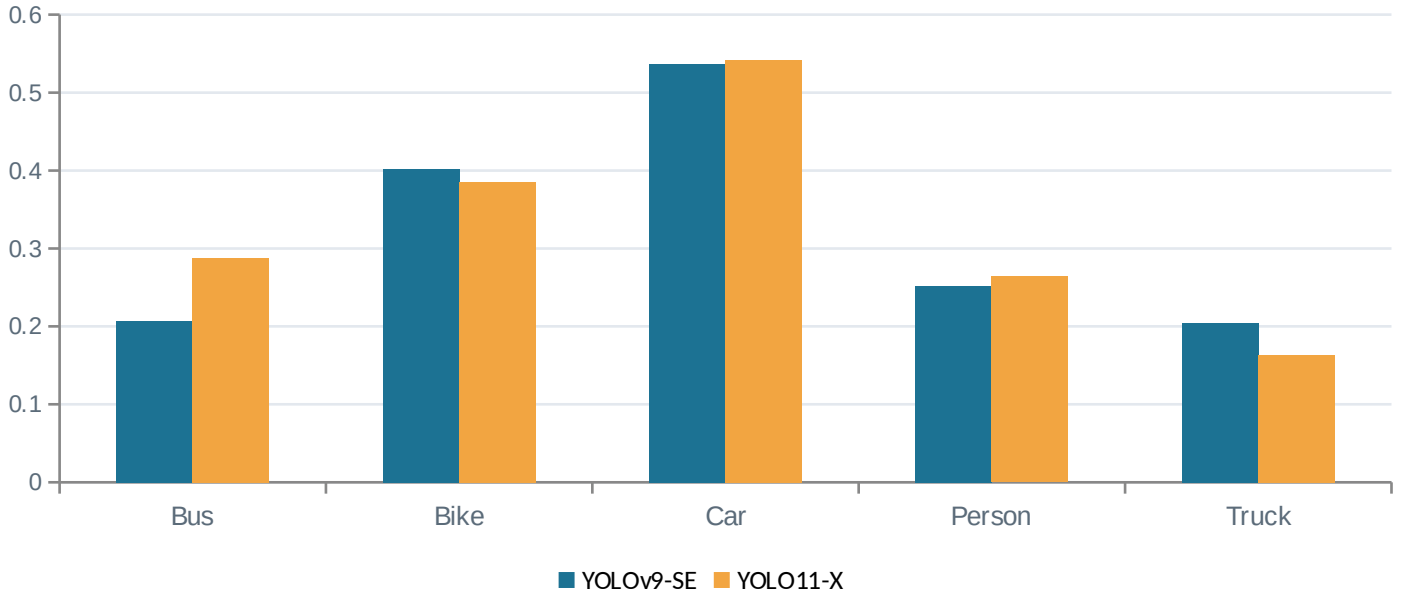
## YOLOv9-SE

Highest F1-Score (0.381) with moderate 7.2 ms inference — best overall trade-off.

**Takeaway:** no single model wins on every axis — the right choice depends on whether the deployment prioritizes latency, precision, or a balanced trade-off between the two.

# Per-Class Detection Performance

**mAP@0.5 by Class — YOLOv9-SE vs. YOLO11-X**




## Key Observations

**Car** is detected most reliably (mAP 0.54, well-represented class).

**Bus & Truck** lag behind — few training instances limit learning.

**Pedestrian** has the lowest recall — small apparent size under fisheye distortion causes frequent misses.

# Pushing YOLO11-X Further

| Configuration | Precision | Recall | mAP@0.5 | mAP@.5:.95 | F1 | Infer (ms) |
|---|---|---|---|---|---|---|
| **100 epochs (baseline)** | 0.502 | 0.297 | 0.329 | 0.166 | 0.373 | 7.1 |
| **200 epochs** | 0.486 | 0.307 | 0.332 | 0.169 | 0.376 | 6.5 |
| **100 + 50 fine-tune + Mosaic** | 0.485 | 0.308 | 0.332 | 0.170 | 0.374 | 6.2 |
| **200 + 50 fine-tune + Mosaic** | 0.475 | 0.280 | 0.318 | 0.164 | 0.347 | 7.4 |

## Observations

- Extending 100 → 200 epochs gives only marginal mAP gains — model is near its plateau.
- Mosaic augmentation fine-tuning nudges recall and mAP@0.5:0.95 up slightly, at a small precision cost.
- Longer training (200 + fine-tune) does not consistently outperform shorter runs — more epochs ≠ better results.

# Confidence & IoU Threshold Tuning

*Adjusting conf = 0.25 and IoU = 0.45 at evaluation time (rather than retraining) unlocked the best accuracy seen in this study.*

| Test Model | mAP@0.5 | mAP@0.5:0.95 |
|---|---|---|
| **YOLO11-X, 100 epochs** | 0.414 | 0.233 |
| **YOLO11-X, 100+50 fine-tune + Mosaic** | **0.419** | **0.238** |
| **YOLO11-X, 200 epochs** | 0.392 | 0.220 |
| **YOLO11-X, 200+50 fine-tune + Mosaic** | 0.403 | 0.228 |

**+0.09**

mAP@0.5 gain
0.329 → 0.419
vs. the original baseline

**Note:** the 100-epoch model with fine-tuning + Mosaic + tuned thresholds outperforms all longer-training variants — confirming that post-training calibration matters as much as training duration.

# Conclusion

**YOLO11-X** delivers the strongest overall accuracy for fisheye road object detection, improving from mAP@0.5 = 0.329 to **0.419** after fine-tuning, Mosaic augmentation, and threshold tuning.

### Best Accuracy

YOLO11-X — highest mAP@0.5 and mAP@0.5:0.95

### Best Balance

YOLOv9-SE — highest F1-Score, moderate speed

### Fastest

YOLO11-L / YOLOv9-C — sub-5ms inference

### Key Insight

Post-training tuning beats longer training alone